CV - Denis PÉRICE

Updated on October 2026



ATER[1] in mathematical physics at Lyon 1 Université Claude Bernard. Read about my research projects.

French, 29 yo perice@math.univ-lyon1.fr https://denis-perice.github.io/

Formation

ENS Lyon - *UMPA*
Lyon, France
Oct 2020 - Oct 2023

Ph.D. in mathematical physics - *Semi-classical limits of 2D fermions under high magnetic fields*
Supervisor: Nicolas Rougerie

ENS Lyon
Lyon, France
2020 - 2021

2nd year master courses in advanced mathematics
Calculus of variations and elliptic equations, kinetic theory

UGA[2] - *UFR-Phitem[3]*
Grenoble, France
2019 - 2020

Master degree in physics - *specialisation in condensed matter physics*
Quantum physics, solid state physics, statistical physics, quantum field theory, relativity, computational physics
Mention good

UGA - *Institut Fourier*
Grenoble, France
2018 - 2020

1st year master courses in fundamentals mathematics
General algebra, differential equations, representation theory, differential geometry, functional analysis, stochastic processes

Grenoble INP[4] - *Phelma[5]*
Grenoble, France
2018 - 2020

Engineering degree - *specialisation in Physics and nano-sciences*
Advanced electromagnetism, semiconductor physics, optical physics, numerical methods, experimental methods in physics and nano-sciences
Mention good

Institut Fourier - *UGA*
Grenoble, France
2017 - 2018

Courses from bachelor in mathematics
Algebra, topology, calculus, measure theory and probability theory

Grenoble INP - *Phelma*
Grenoble, France
2017 - 2018

Bachelor degree - *specialisation in physics, materials, processes*
Quantum mechanics, continuum mechanics, electronic, materials science, statistics, complex analysis, Fourier analysis

CPP[6]
Grenoble, France
2015 - 2017

Polytechnic preparatory cycle - *specialisation in mathematics and mechanic*

Lycée Jean Mermoz
Montpellier, France
2012 - 2015

High school degree in sciences - *specialisation in mathematics*
Mention very good

Scientific schools attended

— Aug 2026, Les Houches, Summer School 2026: Quantum Theory on All Scales
— Jul 2024, Metz, Summer School on Effective Approximation and Dynamics of Many-Body Quantum Systems
— Aug 2022, Venice Quantissima in the Serenissima IV
— Jul 2021, Zürich Summer School on Current Topics in Mathematical Physics
— Aug 2017, Bruxelles Brussels Summer School of Mathematics

Research experience

Université Lyon 1
Lyon, France
Sep 2026 - present

ATER - *Institut Camille Jordan*
Probability, statistics and mathematical physics team
Teaching at Polytech Lyon - Applied mathematics and modelisation track

Constructor university
Bremen, Germany
Nov 2023 - Aug 2026

Postdoc - *DFG[7] grant : Asymptotic Expansions for the Weakly Interacting Bose Gas*
School of sciences
Supervisor: Sören Petrat

CNRS[8] - *UMPA[9]*
Lyon, France
Oct 2020 - Oct 2023

Ph.D. in mathematical physics - *Semi-classical limits of 2D fermions under high magnetic fields*
Modelization and analysis team, ENS Lyon[10]
Supervisor: Nicolas Rougerie

CNRS - *LPMMC[11]*
Grenoble, France
Mar - Jul 2020

Master Internship - *Mean Field limits for fermionic systems under high magnetic field*
Supervisor: Nicolas Rougerie

Osaka university - *Kuwahara lab*
Osaka, Japan
Jun - Aug 2019

Master Internship - *Study of a carbon nanotubes system with a spontaneous modification of conductance for the development of neuromorphic devices*
Supervisor: Megumi Akai-Kasaya

CNRS - *Institut Néel*
Grenoble, France
Jun - Aug 2018

Bachelor internship - *Measurement and analysis of Raman spectroscopy on ceramics, development of python codes for the data treatment*
Supervisor: Denis Testemale

CNRS - *Institut Néel*
Grenoble, France
May - Jun 2017

Bachelor internship - *Synthesis of new multiferroic compounds and characterization with X-ray diffraction*
Supervisor: Céline Darie

Teaching experience

Lecturer - *35h*
Constructor university,
Sep 2025 - May 2026

1st **year Applied Computer Science Bachelor -** *Elements of calculus*
Online class

Lecturer - *35h*
Constructor university,

2nd **year Bachelor of Mathematics -** *Real Analysis*
Measure theory & Fourier analysis

Feb - May 2025

Teaching assistant - *17.5h*
Constructor university,
sep - dec 2024

3rd year Bachelor of Mathematics - *Real analysis*
Topology and functional analysis, supervisor: Shahnaz Farhat

Lecturer - *35h*
Constructor university,
Feb - May 2024

2nd year Bachelor of Mathematics - *Analysis III*
Measure theory & differential calculus

TD[12]/TP[13] - *36h*
Lyon 1 university,
Jan - May 2023

L1[14] Department of mathematics - *Analysis*
TP: Numerical analysis with python
supervisor: Khaled Saleh

TD - *24h*
ENS Lyon,
Sep - Dec 2022

L3[15] Department of mathematics - *Topology and differential calculus*
supervisor: Emmanuel Grenier

Colles[16] - *24h*
Lyon 1 university,
Sep - dec 2022

L2[17] CUPGE[18]
supervisor: Gaëlle Dejou

Lectures/TD - *54h*
Lyon 1 university,
Sep 2021 - Jan 2022

L1 PCSI[19] - *Techniques mathématiques de base (Introduction to analysis)*
supervisor: Ulysse Serres

Colles - *18h*
Lyon 1 university,
Sep - dec 2021

L1 CUPGE
supervisor: Frank Wagner

Publications

— Preprint, with S. Farhat & S. Petrat, Ground state energy of the Bose–Hubbard model with large coordination number with a polaron-type quantum de Finetti theorem
— Ann. Henri Poincaré, with S. Farhat & S. Petrat, Mean field dynamics of the Bose Hubbard model in large dimension
— Anal. PDE (to appear), with N. Rougerie, Gyrokinetic limit of the 2D Hartree equation in a large magnetic field
— J. Math. Phys., Multiple Landau level filling for a large magnetic field limit of 2D fermions
— Ph.D. thesis, Semi-classical limits of 2D fermions under high magnetic fields

Presentations

— Aug 2026, Les Houches, Summer School 2026: Quantum Theory on All Scales, slides
— Jun 2026, Constructor University Bremen, Effective Approximation and Dynamics of Many-Body Quantum Systems, poster, pitch talk slide
— Jun 2026, Universität Kassel, Walkshop on Mathematical Physics, notes
— May 2026, Universität Paderborn, Local Quantum Physics 52 and Mathematical Physics in the Heart of Germany VII, notes

— Oct 2025, Paris, Groupe de travail Physique mathématique, notes
— Oct 2025, Paris-IHP[20], séminaire Problèmes Spectraux en Physique Mathématique, notes
— Sep 2025, Munich, QMATH16 Mathematical Results in Quantum Theory, poster, Springer prize
— Jul 2025, Twente, International Workshop on Operator Theory and its Applications, slides
— May 2025, Oldenburg, 44th Northwest German Functional Analysis Colloquium, notes
— Mar 2025, Rennes, Effective Approximation and Dynamics of Many-Body Quantum Systems Mini Workshop, notes
— Feb 2025, L'Aquila, Mathematical Challenges in Quantum Mechanics school and workshop, slides
— Aug 2024, Bremen, XLIV Dynamics Days Europe, slides
— Jul 2024, Metz, Summer School on Effective Approximation and Dynamics of Many-Body Quantum Systems, poster
— Jul 2024, Strasbourg, XXI International Congress On Mathematical Physics, slides
— Jun 2024, Hagen, Walkshop on Mathematical Physics, slides
— Feb 2024, Braunschweig, Symposium on Mathematical Physics, notes
— Sep 2023, Metz, Kick-off Meeting: Effective Approximation and Dynamics of Many-Body Quantum Systems, notes
— Feb 2023, Bremen, Mathematical and Theoretical Physics Seminar, notes
— Jan 2023, Lyon, UMPA & ICJ[21] - Ph.D. seminar, notes
— Aug 2022, Venice, Quantissima in the Serenissima IV, slides
— Jul 2022, Lyon, UMPA - EDP days, notes
— Jun 2021, Lyon, UMPA - modelization and analysis team seminar, slides
— Feb 2021, Lyon, UMPA - modelization and analysis team seminar, slides
— May 2020, Grenoble, LPMMC weekly seminar, slides

Scientific life

— Jun 2025, Bremen, Organization of the Walkshop in Mathematical Physics
— Aug 2024, Bremen, Technical support in a thematic session for XLIV Dynamics Days Europe

Prizes and others

— Sep 2025, Springer QMATH Poster Session Prize
— Qualification MCF[22] (N°24226402379) valid until 31/12/2028

Glossary

— [1]**ATER:** Attaché Temporaire d'Enseignement et de Recherche (Temporary Teaching and Research Assistant)
— [2]**UGA:** Université Grenoble-Alpes
— [3]**UFR-Phitem:** Unité de Formation et de Recherche en Physique, Ingénierie, Terre, Environnement, Mécanique
— [4]**INP:** Institut National Polytechnique
— [5]**Phelma:** école nationale supérieure de PHysique ELectronique MAtériaux
— [6]**CPP:** Cycle Préparatoire Polytechnique
— [7]**DFG:** Deutsche Forschungsgemeinschaft (German Research Foundation)
— [8]**CNRS:** Centre Nationnal de Recherche Scientifique
— [9]**UMPA:** Unité de Mathématiques Pures et Appliquées
— [10]**ENS Lyon:** École Normale Supérieure de Lyon
— [11]**LPMMC:** Laboratoire de Physique et Modélisation des Milieux Condensés
— [12]**TD:** Travaux Dirigés (Exercice sessions)
— [13]**TP:** Travaux Pratiques (applied exercise sessions)
— [14]**L1:** First bachelor year

— [15]**L3:** Third bachelor year
— [16]**Colles:** Mathematics oral examination in small groups
— [17]**L2:** Second bachelor year
— [18]**CUPGE:** University preparatory cycle for selective french schools
— [19]**PCSI:** Physic Chemistry and engineering sciences
— [20]**IHP:** Institut Henri Poincaré
— [21]**ICJ:** Institut Camille Jordan
— [22]**MCF:** Maître De Conférences